SRINIVAS UNIVERSITY
INSTITUTE OF ENGINEERING & TECHNOLOGY
MUKKA, MANGALURU- 574 146

MICROSERVICES ARCHITECTURE AND IMPLEMENTATION
MINIPROJECT REPORT
ON
FOOD APP

Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

Submitted By,

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2025-2026

DEPARTMENT OF COMPUTER SCIENCE AD ENGINEERING
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CERTIFICATE

This is to certify that the Miniproject report entitled “FOOD APP” is a bonafide work carried out by Deeksha M Naik, Deepa bearing the USN:01SU23CS054, 01SU23CS055 the partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering of the Srinivas University Institute of Engineering and Technology, Mukka Mangalore during the year 2025-2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The Miniproject report has been approved as it satisfies the academic requirements in respect of Miniproject work prescribed for the said degree.

Name & Signature

Prof. Yogesh Bandewar
Miniproject Guide

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DECLARATION

We, Deeksha M Naik, Deepa the student of Sixth semester, B.Tech in Computer Science and Engineering, Srinivas University, Mukka, hereby declare that the Miniproject entitled on "Food App" has been successfully completed by me in partial fulfillment of the requirements for the award of degree in Bachelor of Technology in Computer Science and Engineering of Srinivas University Institute of Engineering and Technology and no part of it has been submitted for the award of degree or diploma in any university or institution previously.

Date: 26-03-2026

Place : Mukka

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ABSTRACT

The Food Ordering Web Application is a web-based system developed to simplify the process of ordering food online. It provides a user-friendly platform where customers can browse a variety of food items, view details such as price and images, and place orders conveniently. The application integrates frontend and backend technologies to ensure smooth interaction between users and the system.

Key features of the system include displaying food menus, adding and removing items from the cart, and managing food data through backend APIs. The application uses a database to store food details, ensuring dynamic content updates. It also handles common issues such as routing errors and data fetching efficiently.

This system is beneficial for both users and developers as it demonstrates real-world implementation of full-stack development. It improves user experience by making food ordering quick and accessible while reducing manual effort. Overall, the Food Ordering Web Application serves as a scalable and efficient solution for online food services.

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CHAPTER 1: INTRODUCTION

1.1 Overview Of The Project

The Food Ordering Web Application is a full-stack web-based system developed to simplify and digitize the process of ordering food. The application provides an interactive platform where users can browse available food items, view details such as price and images, and add items to a cart for ordering. It aims to offer a convenient and efficient alternative to traditional food ordering methods.

The system is built using modern web technologies, combining a frontend interface with a backend server and a database. The frontend allows users to interact with the application, while the backend handles data processing, API requests, and communication with the database. The database stores information about food items, including their names, prices, and images, ensuring dynamic content management.

The application also demonstrates key concepts of full-stack development such as RESTful API integration, client-server architecture, and data handling. Features like adding and removing items from the cart, displaying food menus, and managing food data make the system practical and user-friendly.

Overall, this project provides a foundation for real-world online food ordering systems and can be further enhanced with advanced features such as user authentication, payment integration, and order tracking.

1.2 Problem Definition

In today's fast-paced world, traditional methods of ordering food, such as visiting restaurants or placing orders manually, are time-consuming and inefficient. Customers often face difficulties in viewing available menu items, comparing prices, and placing orders conveniently. Additionally, there is a lack of a centralized system to manage food items, resulting in poor organization and increased chances of errors.

Manual handling of orders can lead to issues such as incorrect order placement, delays in

1.3 Motivation

The motivation behind developing the Food Ordering Web Application arises from the growing need for convenient and efficient online services in daily life. With the increasing use of the internet and smartphones, users expect quick and easy access to food ordering without the need to visit

restaurants physically.

Traditional food ordering methods are often time-consuming, prone to errors, and lack proper organization. These limitations highlight the need for an automated system that can handle food browsing, selection, and ordering in a seamless manner. The project aims to address these challenges by providing a user-friendly interface and efficient backend functionality.

Additionally, this project serves as a practical implementation of full-stack development concepts. It helps in understanding how frontend, backend, and database systems work together to build real-world applications. The motivation also includes enhancing technical skills, gaining hands-on experience, and developing a scalable solution that can be extended with advanced features in the future.

1.4 Objectives

The primary objective of this project is to design and implement an efficient Food Ordering Web Application that allows users to browse food items and place orders online with ease. The system should provide a smooth and interactive experience, enabling users to select food items, add them to a cart, and manage their orders conveniently.

Another objective is to ensure accurate handling of food data and user interactions. The system should properly manage the display of food items, update menus dynamically, and handle user actions such as adding or removing items without errors or delays.

The project also aims to provide a user-friendly interface that simplifies the food ordering process. Users should be able to easily navigate through the application, view food details, and place orders without any complexity.
Additionally, the system should enable efficient backend management of food items through APIs. Administrators or developers should be able to add, update, or delete food data and ensure smooth communication between the frontend and backend.

Finally, the objective is to gain practical experience in full-stack web development by integrating frontend, backend, and database technologies. This helps in enhancing skills in programming, API development, and data management, which are essential for building

CHAPTER 2: LITERATURE REVIEW

2.1 Review and Limitations of Existing Systems

Existing food ordering systems are often based on traditional methods such as in-person ordering at restaurants or placing orders through phone calls. While these methods are commonly used, they come with several challenges. Manual ordering requires more time, may lead to miscommunication between customers and staff, and depends heavily on human effort, making the process less efficient.

The main limitations of these systems include lack of convenience, limited accessibility, and higher chances of errors. Customers must physically visit the restaurant or call to place orders, which can be time-consuming and inconvenient, especially during busy hours. There is also a possibility of incorrect order placement due to misunderstandings.

Additionally, these systems do not provide real-time updates, digital menus, or proper order tracking. Managing records manually becomes difficult, and there is no centralized system to store and analyze data. This reduces efficiency and makes it harder for businesses to handle large numbers of orders effectively.

Therefore, traditional food ordering systems are not well-suited for today's fast-paced digital environment, highlighting the need for an automated and web-based solution.

2.2 Proposed System Overview

The proposed system is a Food Ordering Web Application that allows users to browse and order food online through a web-based platform. It is designed to provide a simple, efficient, and user-friendly experience that can be accessed from anywhere. The system includes features such as displaying food items, viewing details, adding items to a cart, and managing orders through an interactive interface.

The system works by connecting the frontend with a backend server through APIs. When a user accesses the application, food data is fetched from a centralized database and displayed on the screen. Users can select items, add them to the cart, and manage their selections easily. All food-related data is stored and managed in the database, ensuring consistency and easy updates.

CHAPTER 3: SYSTEM ANALYSISANDDESIGN

3.1 Overview

The system architecture for the Food Ordering Web Application is based on a client-server model, consisting of three main components: the user interface (frontend), the application server (backend), and the database. The frontend provides an interface where users can browse food items, view details, and manage their cart. The backend handles the business logic, including data processing, API handling, and communication with the database. The database stores food item details such as name, price, and images securely.

The workflow of the system follows a structured process: users access the application through a web browser, view available food items, and add them to the cart. The frontend sends requests to the backend, which processes the data and retrieves information from the database. Finally, the system displays the selected items and updates the cart accordingly.
User Input → Request to Backend → Data Processing → Database Interaction → Response Display

3.2 Hardware Requirements

- Processor: Intel i3 / i5 / AMD equivalent or higher
- RAM: Minimum 4 GB (8 GB recommended for better performance)
- Hard Disk: Minimum 20 GB free space
- Input Devices: Keyboard and Mouse
- Output Devices: Monitor
- Network: Stable internet connection for accessing the system

3.3 Software Requirements

- Operating System: Windows, Linux, or macOS
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technologies: Node.js / Express.js
- Database: MongoDB for storing food item data

CHAPTER 4: METHODOLOGY

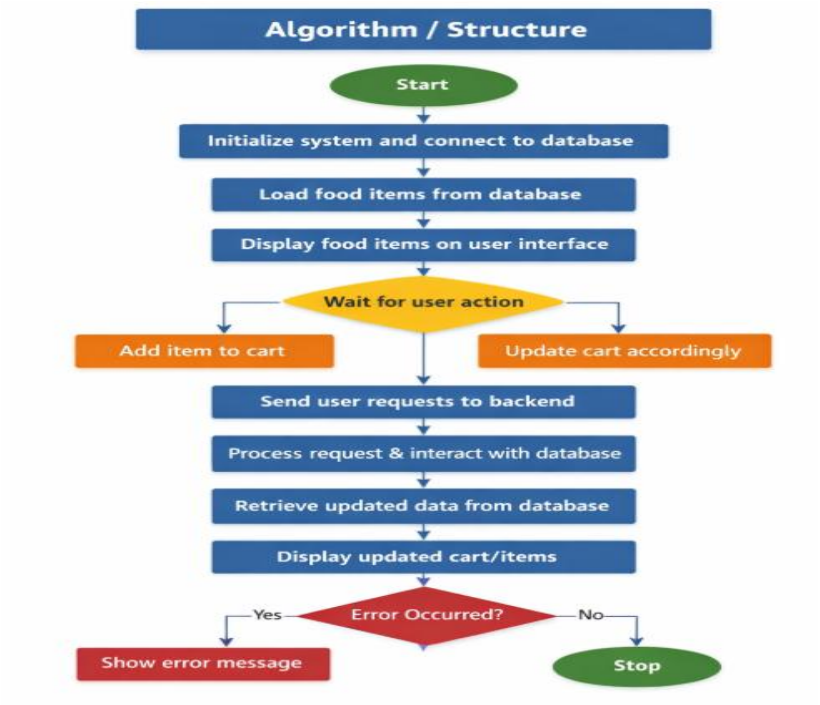
4.1 Working Principle

The Food Ordering Web Application works based on a client-server architecture where users interact with the system through a web interface. Initially, users access the application and browse the available food items displayed on the homepage. The frontend communicates with the backend server to fetch food data from the database.

Once the data is loaded, users can view food details such as name, price, and images, and select items based on their preferences. Selected items can be added to the cart, where users can manage their orders by adding or removing items. The system ensures that all user actions are processed smoothly and updated in real time.

When a user interacts with the application, the frontend sends requests to the backend, which processes the data and performs necessary operations on the database. The updated information is then sent back to the frontend and displayed to the user. The system ensures proper data handling and prevents errors during operations.

4.2 Algorithm

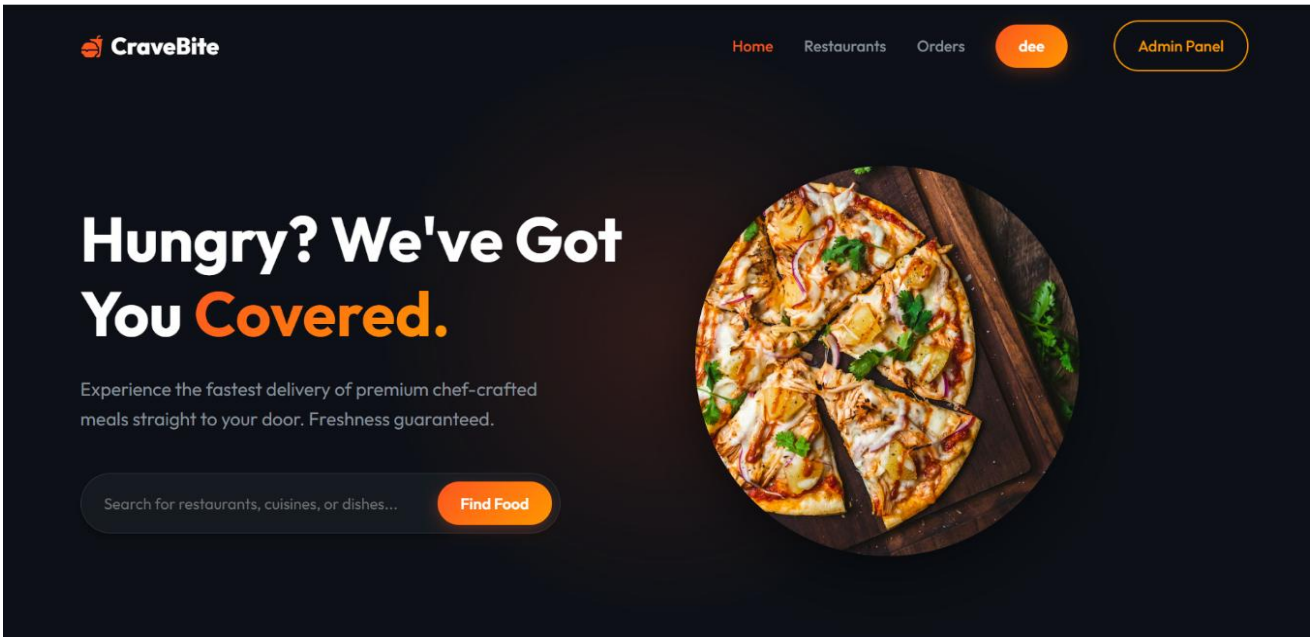


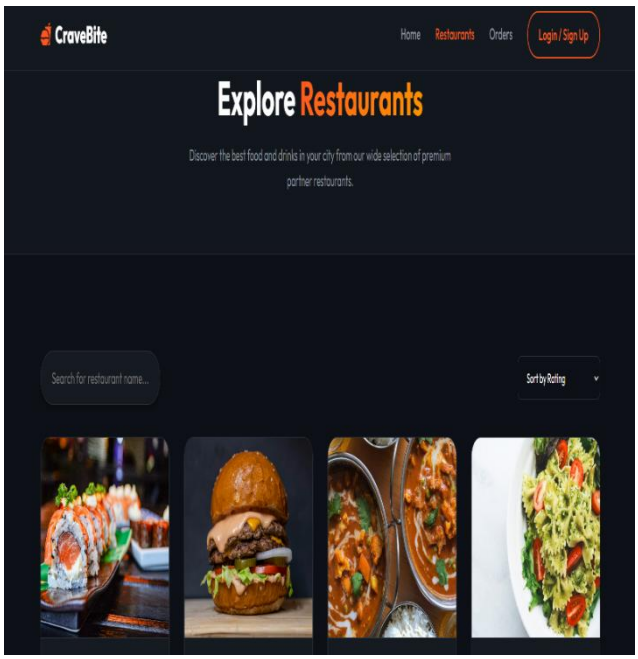
4.3 Module Description

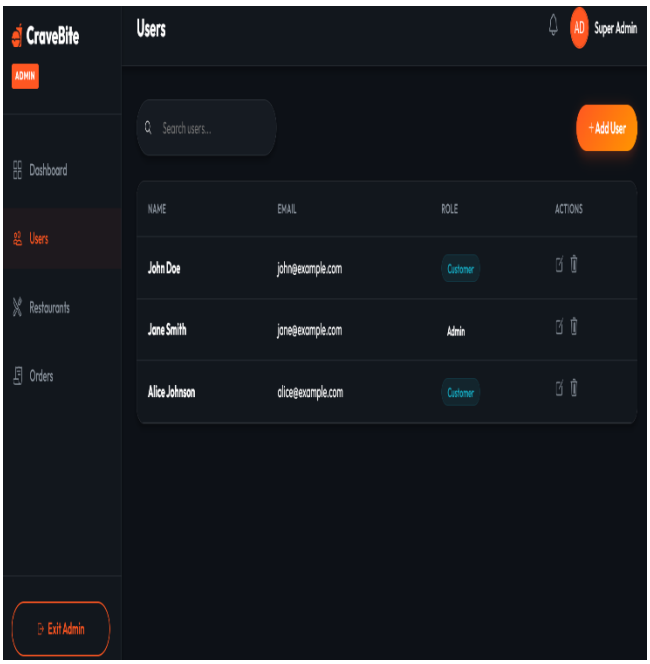
1. **InputModule:**
   Accepts user actions such as selecting food items, adding them to the cart, and interacting with the application interface. It captures user inputs and sends requests to the backend for processing.
2. **AuthenticationModule(Optional):**
   Verifies user identity by checking login credentials (username and password) and ensures that only authorized users can access certain features of the system.
3. **FoodManagementModule:**
   Displays the list of available food items and allows users to view details such as name, price, and images. It also supports backend operations like adding, updating, and deleting food items in the database.
4. **CartModule:**
   Allows users to add food items to the cart, remove items, and manage their selections. It ensures that the cart is updated correctly based on user actions.
5. **OutputModule:**
   Displays results such as updated food lists, cart details, and confirmation messages. It ensures that users receive proper feedback for their actions within the system.

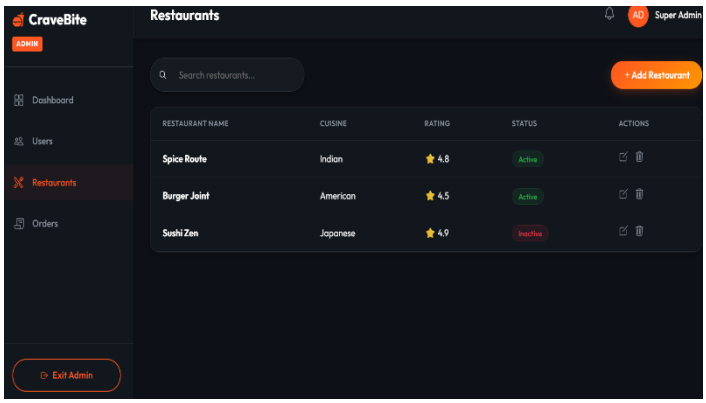
CHAPTER 5: RESULTS

5.1 Output





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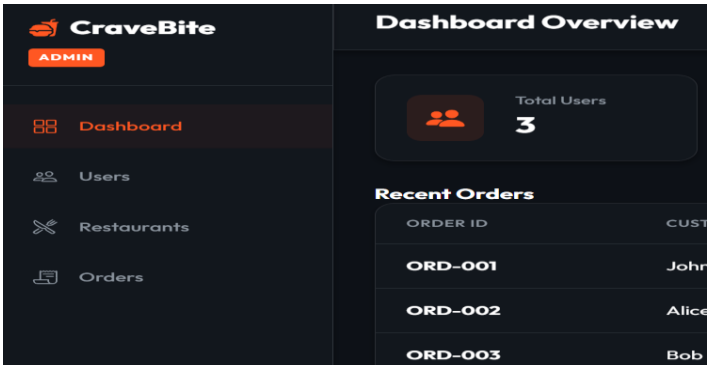
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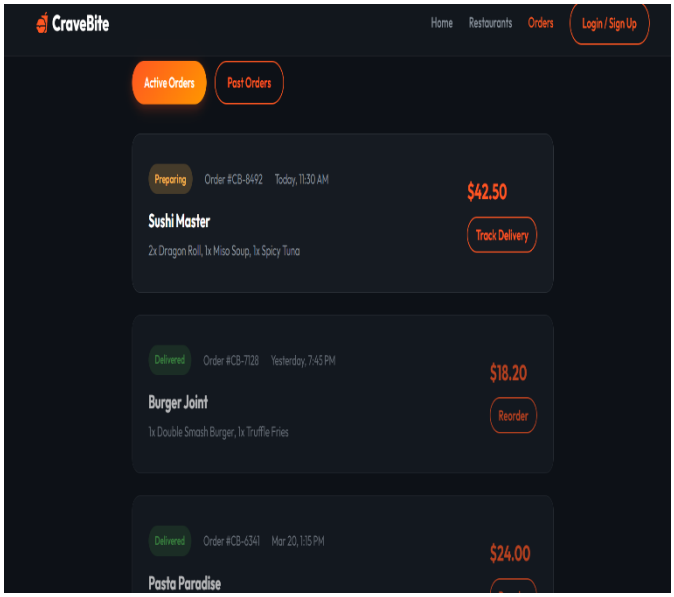
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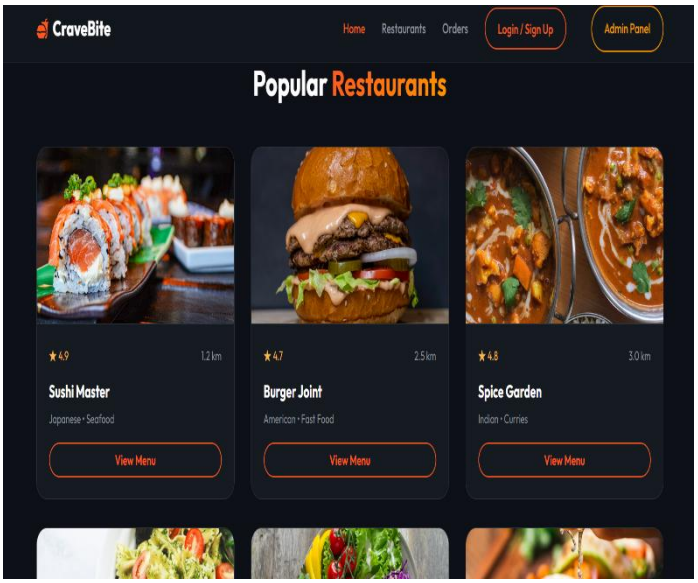
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Sushi Master
Japanese • Seafood
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Burger Joint
American • Fast Food
View Menu
4.8
3.0 km
Spice Garden
Indian • Curries
View Menu

5.2 Challenges

One of the major challenges was handling user interactions efficiently. Ensuring that user actions such as adding and removing items from the cart worked correctly required careful implementation of frontend and backend communication. Initial issues like incorrect updates in the cart were resolved by improving state management and API handling.

Another challenge was managing food data dynamically. It was important to fetch, display, and update food items from the database without errors. Handling this required proper API design and testing to ensure smooth data flow between the server and the user interface.

Database management was also a challenge. Designing the structure to store food item details such as name, price, and images in an organized manner required careful planning. Issues like incorrect queries and data inconsistency were resolved by optimizing database operations.

Handling errors and unexpected user inputs was another difficulty. The system initially produced errors when invalid requests were made or when the backend routes were not properly defined (e.g., “Cannot GET /” error). These issues were resolved by adding proper routing, validation, and error-handling mechanisms.

The system ensures accuracy in order management by reducing human errors that commonly occur in manual ordering. Users can clearly view food details such as price and items before placing their order, minimizing confusion and mistakes.

Ensuring smooth communication between frontend and backend was also challenging. Problems related to API integration and asynchronous requests were addressed through debugging and proper handling of responses.
Finally, testing the application under different scenarios such as multiple user actions and edge cases required extra effort. By thoroughly testing the system, the reliability and performance of the Food Ordering Web Application were improved.

CHAPTER 6: CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusion

The Food Ordering Web Application was successfully designed and implemented as a web-based system to provide an efficient and user-friendly platform for ordering food online. The project demonstrated how modern web technologies and database management can be used to simplify the food ordering process. It allows users to browse food items, manage their cart, and interact with the system smoothly while ensuring accurate data handling.

Moreover, the project provided practical exposure to real-world application development, including frontend design, backend processing, and database integration. It helped in understanding important concepts such as API development, data management, and client-server architecture. By testing the system under different scenarios, its reliability and performance were ensured. Overall, the project enhanced technical skills, problem-solving ability, and knowledge of developing efficient and user-friendly web applications.

6.2 Future Enhancements

In the future, the Food Ordering Web Application can be enhanced by implementing user authentication features such as login and registration to provide personalized experiences. It can also be upgraded to support large-scale usage with improved performance and scalability.

Additional features such as online payment integration, order tracking, and real-time notifications can be added to improve user experience. A mobile application version of the system can also be developed to make food ordering more accessible and convenient. Furthermore, multilingual support can be included to reach a wider range of users.

Advanced features like recommendation systems based on user preferences can be integrated to provide personalized food suggestions. The system can also include admin dashboards, detailed analytics, and reporting tools for better management. Finally, integrating cloud-based infrastructure can improve system availability, reliability, and performance for handling a large number of users efficiently.

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